

Manufacturing Automation Investment Discussion

Date: March 24, 2024

Location: Conference Room / Zoom

Attendees: James Wilson, Sarah Chen, Michael Rodriguez, Emily Thompson

Manufacturing Automation Investment Discussion

Date: [Current Date] Meeting Duration: 90 minutes Attendees: Operations, Finance, Engineering Leadership

Current State Assessment

- Current production capacity: 850 units/month
- Quality metrics: 97.2% first-pass yield
- Labor costs represent 42% of COGS
- Manual assembly operations account for 65% of production hours
- Current bottleneck in final assembly and testing stations

Proposed Automation Solution

Engineering team presented comprehensive automation proposal: - 2 automated assembly cells for primary component integration - Vision-guided robotic testing station - Automated material handling system - Updated MES integration for real-time production monitoring

Capital Investment Requirements

Total Project Cost: \$3.8M - Equipment and Systems: \$2.9M - Infrastructure Updates: \$450K - Software/Integration: \$250K - Training and Implementation: \$200K

Expected Benefits

Production Capacity

- Increase monthly output to 1,200 units (+41%)
- Reduce cycle time by 35%
- Improve OEE from 76% to 89%

Quality Improvements

- Project first-pass yield increase to 99.1%
- Reduce human error-related defects by 80%

- Enhanced traceability through automated data collection
- Consistent process control through automation

Financial Impact

ROI Analysis: - Projected annual savings: \$1.2M - Payback period: 3.2 years - IRR: 28% - Labor cost reduction: \$800K annually - Material waste reduction: \$150K annually

Implementation Plan

Timeline

Phase 1 (Months 1-3): - Facility preparation - Infrastructure updates - Initial equipment ordering

Phase 2 (Months 4-7): - Equipment installation - System integration - Initial testing

Phase 3 (Months 8-10): - Operator training - Production trials - Full implementation

Labor Impact

- Reduction of 12 direct labor positions
- Creation of 4 new technical positions
- Retraining program for 15 existing employees
- New skill requirements in robotics maintenance and programming

Risk Assessment

Primary Risks Identified

1. Production disruption during implementation - Mitigation: Phased implementation during planned downtime
2. Technical integration challenges - Mitigation: Comprehensive testing plan and vendor support agreement
3. Employee resistance - Mitigation: Clear communication plan and retraining opportunities
4. Supply chain delays - Mitigation: Early ordering and alternate vendor identification

Decision and Next Steps

Approved Actions

1. Project approved with \$3.8M budget allocation
2. Engineering to finalize vendor selection by end of month
3. HR to develop detailed retraining program

4. Operations to create detailed implementation schedule

Immediate Next Steps

- Form project steering committee
- Initiate vendor RFQ process
- Begin facility preparation planning
- Develop communication plan for workforce

Success Metrics

- Monthly tracking of:
- Production volume
- Quality metrics
- Labor efficiency
- Cost savings
- Implementation timeline adherence

Next Review: Monthly project status updates to executive team Project Owner: [Operations Director] Project Manager: [Engineering Manager]

Notes compiled by Emily Thompson

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